

Correlation & Regression Techniques

Using MIMIC ICU Data — When to Use Each Method

From association to prediction: a practical, data-grounded guide



Sections: Correlation · Regression · Advanced Methods · Model Selection



Correlation vs. Regression

Correlation — "Are they related?"

- Think of it like a score between -1 and +1
- +1 = both go up together (heart rate & temp during fever)
- -1 = one goes up, other goes down (SpO2 & resp rate)
- 0 = no relationship at all
- Does NOT tell you which one causes the other

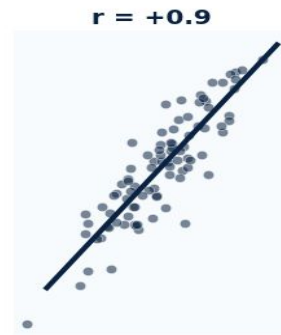
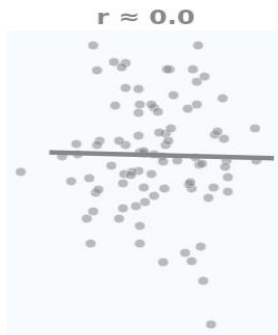
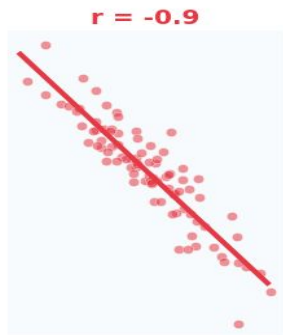
When to use: Use first — before building any prediction model. It's your data scouting trip.

Regression — "Can I predict one from the other?"

- Builds a math formula to predict an outcome
- Output: equation like $LOS = 3.2 + 0.6 \times SOFA$
- Tells you by how much Y changes per unit of X
- You pick the outcome (Y) and the predictors (X)
- MIMIC example: predict ICU stay length from illness score

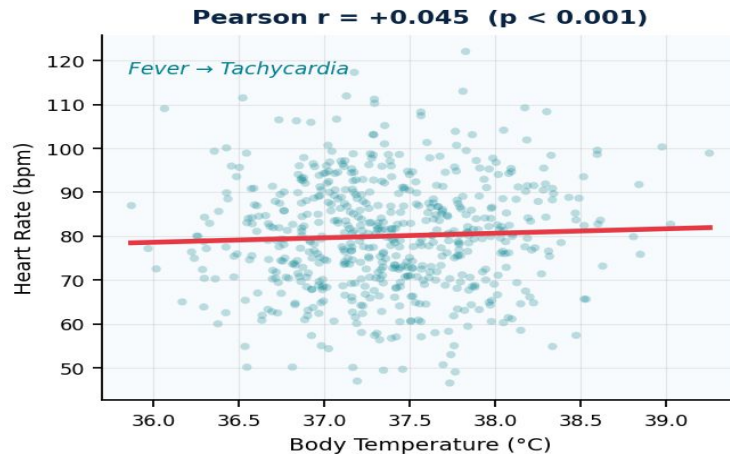
When to use: Use when you need a number: "How many extra days for each SOFA point?"

Pearson Correlation (r) — For Normal, Continuous Data



Summary

- Imagine plotting all patients on a graph — HR on x-axis, Temp on y-axis
- If the dots form a rising line from left to right, r is positive (close to +1)
- If the dots are a random cloud, r is near 0 — no relationship
- Check normality first (Shapiro-Wilk / Q-Q plot) before using Pearson

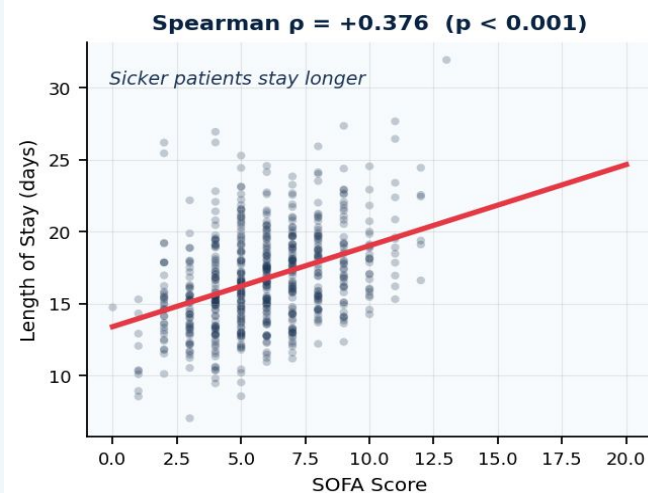


Spearman Correlation (ρ) — When Data is Skewed or Ordinal

The trick: Spearman first converts all your numbers to ranks (1st, 2nd, 3rd...) and **then** runs Pearson on those ranks. This makes it robust to outliers and skew.

	Pearson (r)	Spearman (ρ)
Data type	Continuous, normal	Any — incl. ordinal, skewed
Outlier risk	High	Low (works on ranks)
Relationship	Linear only	Monotonic (can be curved)
MIMIC use	HR vs Body Temp	GCS Score vs LOS

MIMIC tip: LOS is always right-skewed. GCS is ordinal. Use Spearman for both.



Expected: $\rho \approx +0.45$ to $+0.60$ (SOFA vs LOS)

Other Correlation Methods — Kendall, Point-Biserial & sCCA

Kendall's Tau (τ)

What: Compares pairs of patients: do higher ranks on one variable match higher ranks on the other?

When: Small samples | Many tied values | Ordinal data

MIMIC: Severity tiers (mild/moderate/severe) vs number of vasopressors. Lots of ties in severity -> Kendall wins.

Point-Biserial (rpb)

What: Measures the link between a number and a yes/no outcome. It is Pearson applied to a 0/1 variable.

When: Binary outcome (alive/dead, readmitted/not) | Continuous predictor

MIMIC: hospital_expire_flag (0=survived, 1=died) vs SOFA score. Does higher SOFA mean more deaths?

Compare All Three

What: All three give a value between -1 and +1. The difference is how they handle ties, skew, and data types.

When: See the decision guide on slide 12 for a full comparison table

MIMIC: Expected ranges: Kendall tau 0.30-0.50 | Point-biserial 0.40-0.60

Sparse CCA (sCCA)

What: Finds the strongest link between **two groups** of variables at once (e.g. all vitals vs all labs), and automatically selects only the most important ones from each group.

Why use it: Standard correlation handles 2 variables at a time. sCCA handles two entire **blocks** and finds which features in each block drive the correlation — with sparsity keeping results interpretable.

MIMIC: Block 1 = all vitals (HR, Temp, SpO2, RR). Block 2 = all labs (BUN, Creatinine, WBC, Glucose). sCCA finds that HR + Temp from block 1 and BUN + Creatinine from block 2 are the most correlated pair — pointing to a common driver of sepsis severity.

Correlation Pitfalls — What the Number Does NOT Tell You

1 Correlation is NOT Causation

Vasopressor dose correlates with ICU mortality ($r \approx 0.71$). Does that mean vasopressors cause death?

No - both are driven by underlying illness severity (SOFA). Illness severity is the hidden confounding variable.

Fix: Ask yourself 'what third variable could explain this?' Use causal diagrams (DAGs) before interpreting correlation clinically.

2 Spurious Correlations

In MIMIC, number of procedures correlates with mortality. Does performing procedures kill patients?

No - sicker patients receive more procedures AND die more often. Selection bias creates a false signal.

Fix: Stratify your sample by severity before computing correlation. Or use partial correlation to control for the confounder.

3 Restricted Range & Outliers

If you filter MIMIC to only sepsis patients, the HR-Temp correlation may disappear - everyone is already febrile so temperature barely varies. Less variation = weaker correlation.

Fix: Always draw a scatter plot first. Report the min/max range of each variable alongside r so readers can judge whether the result generalises.



Introduction to Regression — Fitting a Line to Noisy Data

Y = What you want to predict

Also called the outcome or dependent variable.

In MIMIC: Length of Stay (days), hospital_expire_flag

X = Your predictors (inputs)

The variables you feed in to make the prediction.

In MIMIC: Age, SOFA score, number of diagnoses, HR

The equation: $y = b_0 + b_1 * x + \text{error}$

b_0 = where the line starts (intercept)

b_1 = slope = how much Y rises for each unit of X

Goal: Ordinary Least Squares (OLS)

The algorithm finds the one line that minimises

total squared distance between the line and the dots

$$y = b_0 + b_1 * x + e$$

MIMIC Example

$$\text{LOS} = 3.1 + 0.6 \times \text{SOFA}$$

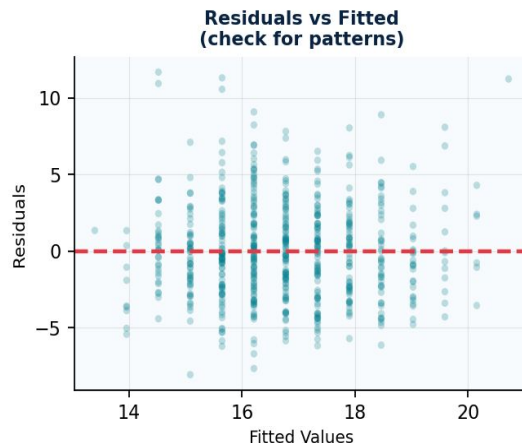
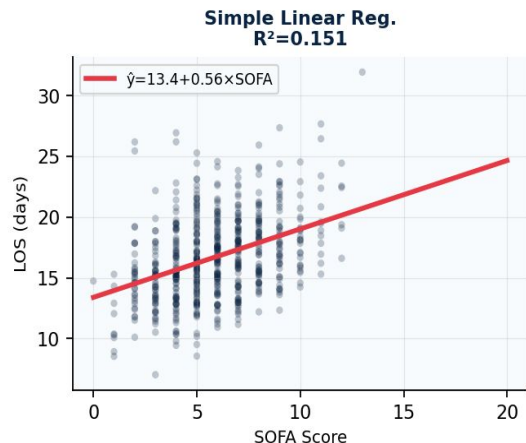
Interpretation:

Each SOFA point adds 0.6 extra days in ICU

b_0 = 3.1 days (baseline LOS when SOFA = 0)

b_1 = 0.6 days per SOFA point

Linear Regression — Simple & Multiple (with Charts)



What the charts show

- Left: scatter + red line = OLS best fit
- Each SOFA point adds ~ 0.6 days to LOS
- Right: residual plot — should be a random cloud around $y=0$
- A pattern in residuals = model assumption violated

Multiple Linear Regression — Adding More Predictors

$$LOS = b_0 + b_1 * \text{Age} + b_2 * \text{SOFA} + b_3 * \text{Comorbidities}$$

Each beta = effect of THAT predictor, holding all others constant

R-squared ~ 0.40 means the three predictors explain 40% of why LOS varies

Assumptions (LINE)

- L = Linearity (scatter should be a line, not a curve)
- I = Independence (patients must be unrelated)
- N = Normal residuals (errors bell-shaped)
- E = Equal spread of errors (homoscedasticity)

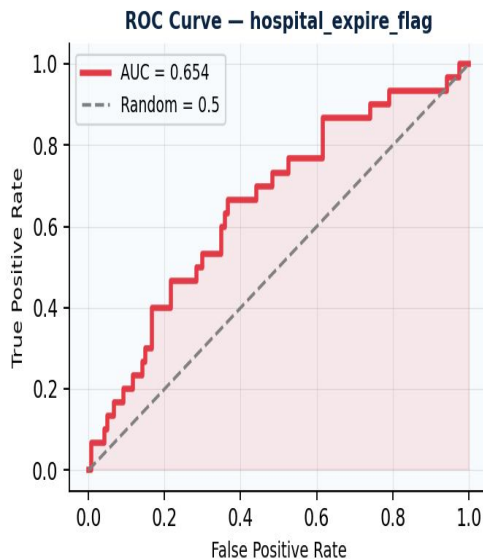
Logistic Regression — Predicting Yes/No Outcomes

Summary

- Goal: predict the PROBABILITY that a patient dies (not a raw number like LOS)
- Output is always between 0 and 1 (0% to 100% chance)
- If predicted probability > 0.5 we classify as high risk (class=1)
- Never use linear regression when your outcome is 0/1 — it can predict probabilities above 1 or below 0, which makes no sense

Sigmoid function: $P = 1 / (1 + \exp(-z))$

- Target: hospital_expire_flag
- Predictors: Age, SOFA, comorbidities, creatinine
- Typical AUC ~ 0.78 to 0.85 in MIMIC studies



Confusion Matrix (threshold=0.5)

Actual \ Predicted	Survived	Died
Survived	118	2
Died	28	2

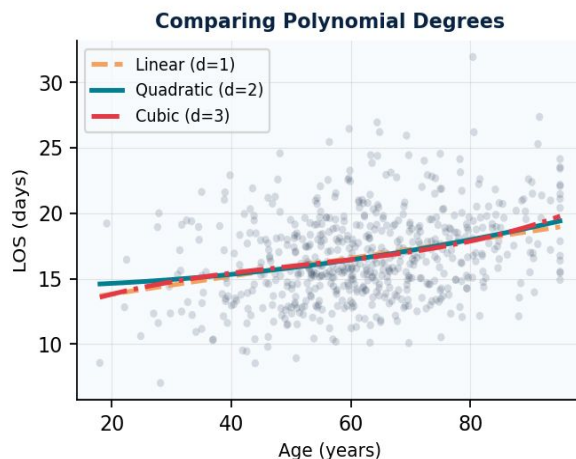
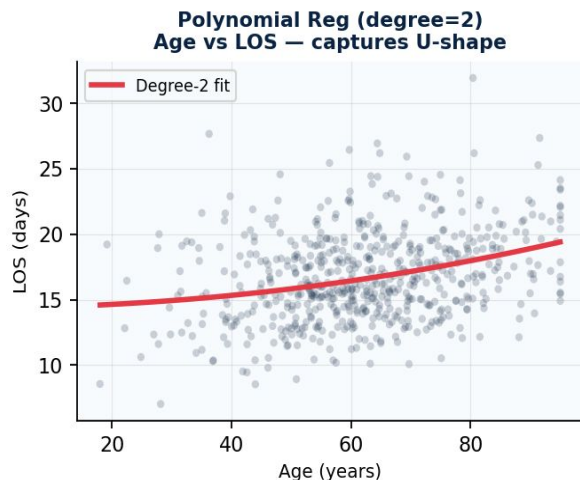
Reading the Confusion Matrix:

Top-left = True Negatives (correctly predicted survival) Bottom-right = True Positives (correctly predicted death)

Off-diagonal cells are errors: False Positives = alarm with no death | False Negatives = missed death

Polynomial Regression — When the Relationship Is a Curve

Key insight: A polynomial model is still a **linear** model — it just adds x^2 , x^3 as extra columns. Same OLS math, curved result.



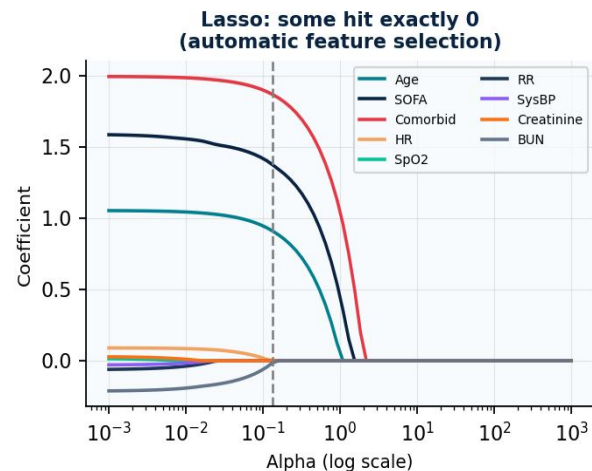
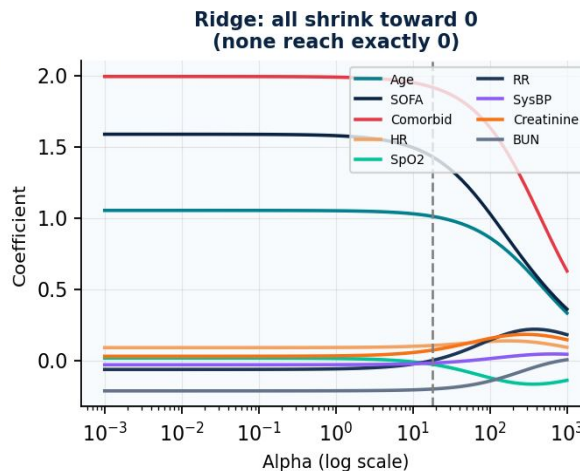
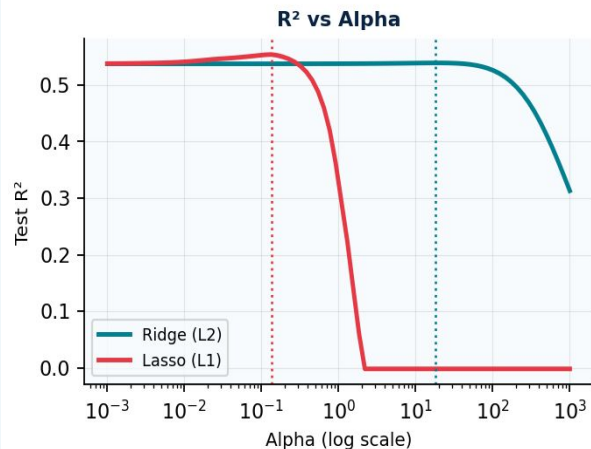
When to use

- Scatter shows a curve, not a straight line
- Biological U-shape (e.g. age vs LOS — very young AND very old leave sooner)
- Linear residuals show a curved pattern
- Warning: degree 3+ almost always overfits — use cross-validation to choose degree

MIMIC Example: Age vs LOS. Younger patients recover faster. Very elderly patients (>85) often have shorter stays due to palliative goals. A degree-2 (quadratic) polynomial captures this arch shape that a straight line completely misses.



Advanced Regression — Ridge, Lasso & Decision Trees (with Charts)



Ridge (L2)

Shrinks ALL coefficients toward zero but none reach zero. Good when all predictors matter. Fixes instability from correlated variables.

MIMIC: All vitals -> LOS. HR, Temp, SpO2, RR are correlated. Ridge stabilises the model.

Lasso (L1)

Some coefficients become EXACTLY zero. Automatic feature selection. Use when you suspect many predictors are irrelevant.

MIMIC: 30+ Elixhauser flags -> readmission. Lasso picks the relevant 5-8 and ignores the rest.

Decision Trees

No linearity needed. Produces simple IF-THEN rules. Captures threshold effects like 'creatinine > 2 mg/dL changes everything'.

MIMIC: Discharge vitals -> 48h ICU readmission. Tree: IF HR > 100 AND SpO2 < 93 THEN HIGH RISK.



Choosing the Right Method — Decision Matrix

Goal	Outcome type	Data conditions	Method	MIMIC example
Describe association	Both continuous	Normal, no outliers	Pearson (r)	HR vs Body Temp
Describe association	Continuous / ordinal	Skewed, ordinal, or outliers	Spearman (ρ)	GCS vs LOS
Association (small N)	Both ordinal with ties	Small sample, many ties	Kendall's Tau	Severity tier vs vasopressors
Association	Continuous vs binary	Binary is naturally 0/1	Point-Biserial	SOFA vs hospital_expire_flag
Multi-block association	Two sets of variables	High-dimensional, need sparsity	sCCA	Vitals block vs Labs block
Predict continuous Y	Continuous	One predictor, linear	Simple Linear Reg.	SOFA → LOS
Predict continuous Y	Continuous	Multiple predictors	Multiple Linear Reg.	Age+SOFA+Comorbid → LOS
Predict probability	Binary (0 or 1)	Any	Logistic Regression	hospital_expire_flag
Handle non-linearity	Continuous	Curved scatter plot	Polynomial Regression	Age → LOS (U-shape)
Correlated features	Continuous	All features matter	Ridge (L2)	All vitals → LOS
Feature selection	Continuous / binary	Many irrelevant features	Lasso (L1)	Elixhauser flags → readmission
Non-linear, mixed types	Continuous / binary	Threshold effects present	Decision Tree	Vitals → ICU readmission



Core Takeaways

- 1 Explore with correlation first — Pearson (normal data), Spearman (skewed/ordinal), sCCA (two variable blocks).
- 2 Match your method to your outcome: continuous → Linear, binary → Logistic, curved → Polynomial.
- 3 Ridge when predictors are correlated and all matter; Lasso when you need to select from many features.
- 4 Always check assumptions and plot residuals — a high R^2 with violated assumptions is unreliable.
- 5 Correlation never proves causation. Every time you find an association, ask: 'what confounder could explain this?'

"Correlation finds the pattern. Regression models it. Good judgment decides what it means."